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journal homepage: www.elsevier.com/locate/saaBiological approach to synthesize TiO_2 nanoparticles using *Aeromonas hydrophila* and its antibacterial activity

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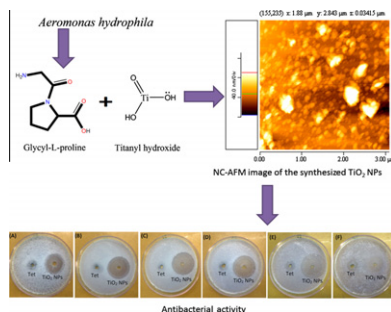
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HIGHLIGHTS

- *Aeromonas hydrophila* mediated biosynthesis of TiO_2 NPs was achieved.
- Glycyl-L-proline plays an important role as a capping and stabilizing agent.
- XRD confirmed the crystalline nanoparticles of 40.50 nm in size.
- FESEM analysis showed smooth in shape, spherical and uneven.
- TiO_2 NPs exposed great inhibition zone in *S. aureus* (33 mm) and *S. pyogenes* (31 mm).

GRAPHICAL ABSTRACT

Development of reliable and eco-friendly process for synthesis of metal oxide nanoparticles is an important step in the field of application of nanotechnology. Titanium dioxide (TiO_2 NPs) nanoparticles were synthesized using the bacteria *Aeromonas hydrophila*. The synthesized particles were subjected to characterization by FTIR, XRD, FESEM and AFM. Non-contact mode AFM images of the TiO_2 NPs find the topological appearance and the size. The range was found to be 28–54 and the individual particle size was 40.50 nm.



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ABSTRACT

Nanosized materials have been an important subject in basic and applied sciences. A novel, low-cost, green and reproducible bacteria, *Aeromonas hydrophila* mediated biosynthesis of titanium dioxide nanoparticles (TiO_2 NPs) was reported. The resulting nanoparticles were characterized by FTIR, XRD, AFM and FESEM with EDX. FTIR showed characteristic bands (1643 and 3430 cm^{-1}) finds the role of carboxyl group O–H stretching amine N–H stretch in the formation of TiO_2 NPs. The XRD spectrum confirmed that the synthesized TiO_2 NPs were in the form of nanocrystals, as evidenced by the peaks at 2θ values of 27.47° , 31.77° , 36.11° , 41.25° , 54.39° , 56.64° and 69.54° were identified as 110, 100, 101, 111, 211, 220 and 301 reflections, respectively. The crystallite sizes were calculated using Scherrer's formula applied to the major intense peaks and found to be the size of 40.50 nm. The morphological characterization was analyzed by FESEM and the analysis showed the NPs smooth shaped, spherical and uneven. GC–MS analysis showed the main compounds found in *A. hydrophila* were uric acid (2.95%), glycyl-L-glutamic acid (6.90%), glycyl-L-proline (74.41%) and L-Leucyl-D-leucine (15.74%). The potential glycyl-L-proline could have played an important role as a capping agent. A possible mechanism for the biosynthesis of

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TiO₂ NPs has been proposed. The antibacterial activity of the synthesized TiO₂ NPs was assessed by well diffusion method toward *A. hydrophila*, *Escherichia coli*, *Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Streptococcus pyogenes* and *Enterococcus faecalis* and showed effective inhibitory activity against *S. aureus* (33 mm) and *S. pyogenes* (31 mm).

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Introduction

The synthesis and assembly of nanoparticles would benefit from the development of clean, nontoxic and environmentally acceptable green chemistry procedures, probably involving organisms ranging from bacteria to fungi and even plants [1]. The use of highly structured physical and biosynthetic activities of microbial cells for the synthesis of nanosized materials has recently emerged as a novel approach for the synthesis of the metal nanoparticles [2]. Among several oxide semiconductors, TiO₂ has a more helpful role in our environmental purification due to its non-toxicity, photo catalytic activity, photo induced super-hydrophilicity and anti-fogging effect. These properties have been applied in removing bacteria and harmful organic materials from water and air, as well as in self-cleaning or self-sterilizing surfaces in medical centers [3]. TiO₂ is widely used as an ingredient in white paint and building materials, food colorant, sunscreen and cosmetic products [4]. The opacity and whitening properties of TiO₂ NPs have a variety of industrial applications, including the manufacture of paints, textiles, papers, plastics, sunscreens, cosmetics, and food products [5]. Suspended TiO₂ NPs have been largely used as efficient catalyst for the decomposition of organic contaminants present in water and aqueous wastes [6,7]. Recent studies on the use of microorganisms in the synthesis of nanoparticles are relatively new and exciting area of research with considerable potential for development. The recent discovery of biosynthesis of metal nanoparticles points towards new biotechnological methods in materials science [8,9].

The non-toxic and biocompatible properties of titania find its applications in biomedical sciences such as bone tissue engineering as well as in pharmaceutical industries [10]. The biomass of *Aeromonas* SH10 was proven to strongly absorb Ag⁺ and [Ag(NH₃)₂]⁺ and the FTIR spectra analysis indicated that some organic groups, such as amide and ionized carboxyl in the cell wall, played an important role in the process of biosorption [11,12]. *Lactobacillus* sp. and *Saccharomyces cerevisiae* have been taken in order to assess its potential as putative candidate bacterial and fungal genera for the synthesis of TiO₂ NPs [13]. Recently, green TiO₂ NPs have been synthesized using natural products like *Nyctanthes arbortristis* extract [14]. Bansal et al. [15] reported the fungus (*Fusarium oxysporum*) mediated synthesis of TiO₂ NPs. Extracellular products (ECPs) of *Aeromonas hydrophila* as microbial antigen was efficiently loaded to form Oleoyl-carboxymethyl chitosan-ECPs nanoparticles and shown to be sustained release in PBS [16].

The antibacterial activity of TiO₂ has been found to be due to a reaction of the TiO₂ surface with water. On exposure to ultraviolet irradiation, TiO₂ releases free radicals such as OH, O₂[•], HO₂[•], and H₂O₂. This potent oxidizing power characteristically results in case of bacteria and other organic substances [17]. Nanoscale particles have emerged as novel antimicrobial agents owing to the high surface area to volume ratio, which was coming up as the current interest for researchers due to the growing microbial resistances against metal ions, antibiotics and the development of resistant strains.

Metal oxide nanoparticles, especially TiO₂ and Ag₂O nanoparticles have demonstrated significant antibacterial and antileishmanial effects [18]. Glass-ionomer containing 3% (w/w) TiO₂ NPs behaved as a promising restorative material with improved mechanical and antibacterial properties [19]. The antibacterial

properties of nanoparticles were investigated by agar diffusion method toward *Escherichia coli* and *Bacillus subtilis* and the results indicated that both Ag and N-doped TiO₂ could increase the antibacterial properties of TiO₂ NPs under fluorescent light irradiation [20]. TiO₂ bactericidal activity starts its action by oxidative damage to the bacteria wall; a decrease in the interfacial energy of bacteria adhesion causes an increase in the chemical interaction between *E. coli* and the diamond-like carbon films, which is an additional factor for the increase in bactericidal activity [21]. The composite TiO₂–Ag coating showed complete killing of methicillin-resistant *Staphylococcus aureus* within 24 h in all culture conditions, whereas a 1000-fold increase in bacterial numbers was recorded with the ground titanium specimens and the samples oxidized in the absence of Ag nanoparticles [22]. Visible-light-activated TiO₂ indicated that a significantly higher proportion of all tested pathogens including *S. aureus*, *Shigella flexneri* and *Acinetobacter baumannii* were eliminated by the nanoparticle with higher bacterial interaction property [23]. Reactive oxygen species, such as [•]OH, O₂[•], and H₂O₂ generated on the light irradiated TiO₂ surfaces were shown to operate in concert to attack polyunsaturated phospholipids in bacteria [24]. Antibacterial activities of apatite-coated TiO₂ against *S. aureus*, *E. coli*, methicillin-resistant *S. aureus* (MRSA), and *Micrococcus luteus* were investigated and suggested for its potential use in reducing the risk of microorganism transmission in textile applications [25].

The use of toxic chemicals on the surface of nanoparticles and non-polar solvents in the synthesis procedure limits their applications in clinical fields. Herein, we report a simple biological method for the synthesis of TiO₂ NPs using bacteria *A. hydrophila* and its antibacterial activity.

Materials and methods

Materials

A. hydrophila (MTCC-1739), *E. coli* (MTCC-1677), *Pseudomonas aeruginosa* (MTCC-4030), *Streptococcus pyogenes* (MTCC-1928), *S. aureus* (MTCC-3160), and *Enterococcus faecalis* (MTCC-3159) were obtained from Institute of Microbial Technology (IMTECH), Chandigarh, India. All the media and chemicals were purchased from Himedia Laboratories Pvt. Ltd., Mumbai, India.

Synthesis of TiO₂ NPs

Bacterial culture was grown in a modified nutrient broth (beef extract 1%, peptone 1%, NaCl 0.5% and pH 7) at 30 °C in an orbital shaker set at 120 rpm for 24 h. 25 ml of culture was taken and diluted four times by adding 75 ml of sterile distilled water containing nutrients. This diluted culture solution was again allowed to grow for another 24 h. 20 ml of 5 mM TiO(OH)₂ was added to the culture solution and kept under shaking incubator at 120 rpm at 30 °C for 24 h until white deposition starts to appear at the bottom of the flask, indicating the initiation of transformation. At the end of the reaction time (24 h) the mixture was poured on a sterilized petriplates and air dried. The air dried powder nanoparticles were used for further analytical studies [26–28].

Analysis of secondary metabolites

Diluted culture broth composition was investigated using a GC–MS GCD-HP1800A system (Hewlett–Packard, USA) equipped with a split/splitless capillary injection port. For GC–MS detection, an electron ionization system (Quadrupoles Analyzer – Mass range: 10–425 amu) with ionization energy of 70 eV was used. The composition of dilute culture broth were identified after comparison with those available in the computer library (NIST) attached to the GC–MS instrument and reported.

Characterization of TiO₂ NPs

Characterization involved FTIR analysis of the dried powder of synthesized TiO₂ NPs, by scanning it in the range 350–4000 cm^{−1} at a resolution of 4 cm^{−1}. These measurements were carried out on a Perkin Elmer Spectrum One instrument in the diffuse reflectance mode at a resolution of 4 cm^{−1} in KBr pellets. For comparison, a drop of 20% bacterial synthesized TiO₂ NPs powder was mixed with KBr powder and pelletized after drying properly.

The formation and quality of compounds were checked by XRD technique. The purified powders obtained after 1 h of interaction under laboratory conditions was subjected to X-ray diffraction analysis. The spectra were recorded in Bruker AXS D8 Advance X-ray diffractometer with Philips® PW 1830 X-ray generator. The generator was operated at 40 kV and with a 30 mA current. The scanning range was selected between 10° and 100°.

Topography was studied using an atomic force microscope (AFM) (Veeco Pico Force) working in the contact mode. In particular, AFM were used to characterize the uniformity and grain size of TiO₂ films deposited on different substrates. AFM images have been processed using WSxM software ver. 4.0.

The FESEM analysis was based upon a beam of high energy electrons generating a variety of signals at the surface of solid specimens. The signals that derive from electron sample interactions revealed external morphology (texture), chemical composition, crystalline structure and orientation of materials making up the sample. In most applications, data were collected over a selected area of the surface of the sample, and two dimensional images

were generated which displayed spatial variations in these properties. The nanoparticle samples were ion-sputtered to avoid charge effect with 2–3 nm carbon layers using a Gatan 681 high resolution ion beam coat at an angle of 45°. Argon ion beam sputter was operated at 6 mA at an acceleration voltage of 15 kV. Secondary electron images were obtained in FESEM JSM-6700, JEOL, Japan at an accelerating voltage of 5.0 kV at 12 Å emissions current and work distance at 8.0 mm.

Antibacterial activity

Well diffusion technique

The antibacterial activity of the synthesized NPs was carried out by Well diffusion method. The obtained bacterial cultures were maintained on nutrient agar slants that contained peptone, 5.0; meat extract, 1.0; yeast extract 2.0; sodium chloride, 5.0; and agar 15.0 g per liter of distilled water. The test bacterial suspensions (50 µl) containing 10⁴ cells ml^{−1} were spread on Nutrient Agar plates. 25 µl TiO₂ NPs suspension (25 µg) was added in the tested wells. Tetracycline 25 µl used in well diffusion technique served as positive control. The samples were initially incubated for 15 min at 4 °C (allow diffusion) and later on at 37 °C for 24 h for the bacterial cultures. Positive test results were scored when zone of inhibition was observed around the well after the incubation period.

Minimum inhibitory concentration (MIC)

The microdilution method for estimation of MIC values was carried out to evaluate the antimicrobial activity. The MIC values were determined on 96-well microdilution plates according to published protocols [29].

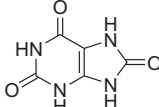
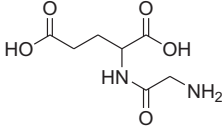
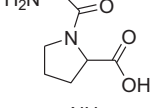
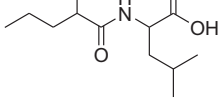
Results and discussion

Identification of secondary metabolites

Compositions of diluted broth culture were identified using NIST library attached to the GC–MS instrument. The major secondary metabolite which is present in the culture broth is glycyl-L-proline (74.41%). The molecular formula of glycyl-L-proline is

Table 1

Compositions of diluted broth culture were identified using NIST library attached to the GC–MS instrument.

Peak	<i>t_r</i> (min)	<i>m/z</i> / [<i>M</i> + <i>H</i>] ⁺	Identified molecule	Chemical structure	% Peak
A	3.1	689	Uric acid		2.95
B	9.6	637	Glycyl-L-glutamic acid		6.90
C	10.1	855	Glycyl-L-proline		74.41
D	20.3	619	L-Leucyl-D-leucine		15.74

The major compound which is present in the extract is glycyl-L-proline. Therefore it is the responsible compound for the formation of TiO₂ nanoparticles.

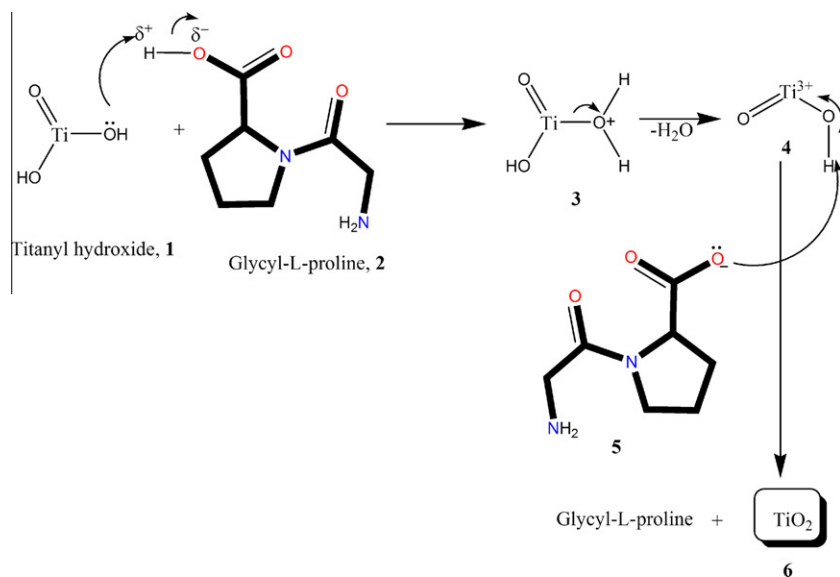


Fig. 1. Possible mechanism for the synthesis of nanosize TiO_2 particles.

$\text{C}_7\text{H}_{12}\text{N}_2\text{O}_3$ and the molecular weight is 172. We have observed the M^+ peak (i.e., 172) in the mass report; it is clearly supporting the compound of glycyl-L-proline. The other minor constituents are uric acid (2.95%), glycyl-L-glutamic acid (6.90%) and L-Leucyl-D-leucine (15.74%) and the major compounds contains $-\text{COOH}$ and $-\text{C}=\text{O}$ group (Table 1). Hence, they may play an important role in the synthesis of TiO_2 nanoparticles as capping and stabilizing agents. Thus we may safely conclude that the glycyl-L-proline of *A. hydrophila* acts as reducing, stabilizing, and capping agents.

Possible mechanism for the synthesis of nanosize TiO_2 particles

In order to find out the role of secondary metabolites which is present in *A. hydrophila* broth, GC-MS has been evaluated. It shows that the broth extract contains compounds having acid group carboxylic acid as a functional group in the structure. The major metabolite which is present in the broth is glycyl-L-proline. Titanyl hydroxide can be dehydrated to give TiO_2 nanoparticles by reacting with an *A. hydrophila* broth at about 30°C for 24 h. In the first stage, one of the lone pairs of e^- on the oxygen, (1) picks up a hydrogen ion from glycyl-L-proline, (2) the $\text{TiO}(\text{OH})_2$ is said to be protonated, (3) the protonated $\text{TiO}(\text{OH})_2$ loses a water molecule to give Ti^{3+} , (4) finally, compound, (5) pulls off a hydrogen ion from the Ti^{3+} . Hence the reaction between the broth of *A. hydrophila* aqueous and $\text{TiO}(\text{OH})_2$ might occur through the possible pathway which is described in Fig. 1. Therefore, water soluble compounds containing carboxylic acid group are reported to be responsible for the stabilization of TiO_2 nanoparticles.

FTIR analysis

The FTIR for $\text{TiO}(\text{OH})_2$ and TiO_2 NPs synthesized by *A. hydrophila* are shown in Fig. 2A and B respectively. The FTIR spectrum of the $\text{TiO}(\text{OH})_2$ showed characteristic bands at 3430 and 1643 cm^{-1} which correspond to the surface water and hydroxyl group. The band intensities of the spectrum for the synthesized TiO_2 NPs are 3430, 2937, 1643, 1403 and 1079 cm^{-1} . These results indicated that alcohols, phenols, primary amines, lactones and aliphatic amines in *A. hydrophila* may have been participated in the process

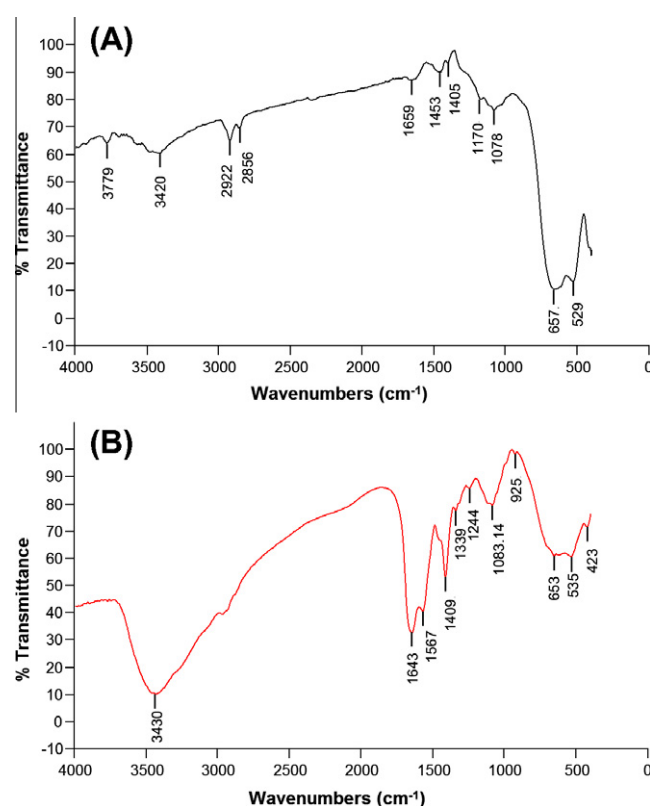


Fig. 2. FTIR spectrum of: (A) $\text{TiO}(\text{OH})_2$ and (B) synthesized TiO_2 NPs.

of nanoparticle synthesis. Functional groups associated with these were the cause for the bioreduction of $\text{TiO}(\text{OH})_2$ to TiO_2 NPs [30].

X-ray diffraction analysis

The crystalline nature of the TiO_2 NPs was analyzed using XRD. The mean particle diameter of the nanoparticles was calculated from the XRD pattern using the Scherrer equation:

$$D = K\lambda / \beta_{1/2} \cos \theta$$

K is the shape constant λ is the wavelength of the X-ray, $\beta_{1/2}$ and θ are the half width of the peak and half of the Bragg angle, respectively. XRD pattern for the control sample [TiO(OH)₂] (Fig. 3A) and synthesized TiO₂ NPs (Fig. 3B) were recorded and the data were compared. The XRD pattern of the synthesized TiO₂ NPs revealed intense peaks at 27.47°, 31.77°, 36.11°, 41.25°, 54.39°, 56.64° and 69.53° corresponding to 110, 100, 101, 111, 211, 220 and 301 Bragg's reflection, respectively. Which is found to be that of the rutile form as when compared with the JCPDS data (File No. 99-101-0954). The main peak of $2\theta = 27.47^\circ$ matches the (110) crystallographic plane of rutile form of TiO₂ nanoparticles, indicating that the nanoparticles structure produced is only rutile form and not of anatase or brookite form. The crystallite sizes were calculated using Scherrer's formula applied to the major intense peaks and found to be in the range of 40.50 nm. The broadening of the peaks clearly shows the small size of the nanoparticles [31,32].

AFM studies

The synthesized TiO₂ NPs were characterized by AFM for its detail size, morphology and agglomeration. Characterization of the synthesized nanoparticles using AFM offered a three-dimensional visualization. The uneven surface morphology was explained by the presence of both individual and agglomerated nanoparticles. The strong crystalline nature can be seen in the form of diagonal formations with ridges (Fig. 4A and B). The AFM provides suitable data regarding the biological preparation and the application of *A. hydrophila* in the synthesis of the metal based nanoparticles. In accordance with the above results, the same influence of TiO₂ concentration and crystallization temperature was seen on the roughness of the surface of TiO₂. By means of AFM no linear trend in roughness was observed, but it is proved that the highest TiO₂ concentration result in the formation of smoother layers [33].

FESEM analysis

The surface morphology of nanoparticles was investigated using FESEM. The nanoparticles were distributed uniformly on the surface with formation of aggregated nanoparticles. It shows that the nanoparticles were densely dispersed with a narrow range of dispersion. Particles were of nanosize with smooth surface. Fig. 5A and B shows that the size of the TiO₂ NPs was very consistent. EDX showing chemical composition and purity of the TiO₂ NPs (Fig. 5C). Fig. 5D shows the particle size distribution. The particle size of the nanoparticles ranges in size from 28 to 54 nm and the individual size of 40.50 nm. The results indicate that these surface planes having different packing structures were energetically similar to each other. The formation of TiO₂ NPs synthesized from *A. hydrophila* with two insets showing the spherical and uneven nanoparticles. The surface of TiO₂ NPs was analyzed and the aggregated structures possess considerable surface roughness. The morphological shapes and size of the aggregations were described in terms of the fractal dimensions and box-counting method [34].

Well diffusion and minimum inhibitory concentration

In the antibacterial activity, synthesized TiO₂ nanoparticles showed maximum zone of inhibition against *S. aureus* (33 mm) followed by *E. coli* (26 mm), *P. aeruginosa* (25 mm), *S. pyogenes* (31 mm), *E. faecalis* (29 mm), and the least was noticed against *A. hydrophila* (23 mm). Positive control of tetracycline showed maximum zone of inhibition against *E. faecalis* (16 mm) and the least was noticed against *P. aeruginosa* (12 mm) (Fig. 6 and Table 2).

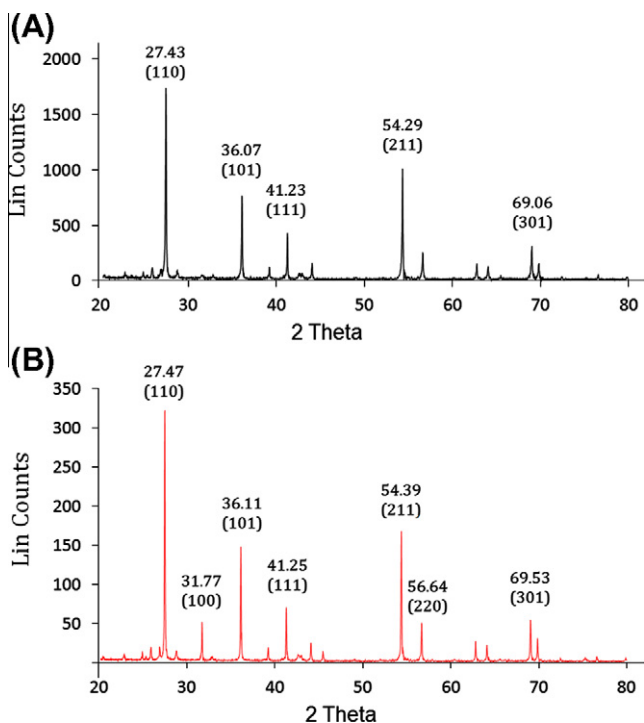


Fig. 3. X-ray diffraction patterns of: (A) TiO(OH)₂ and (B) synthesized TiO₂ NPs.

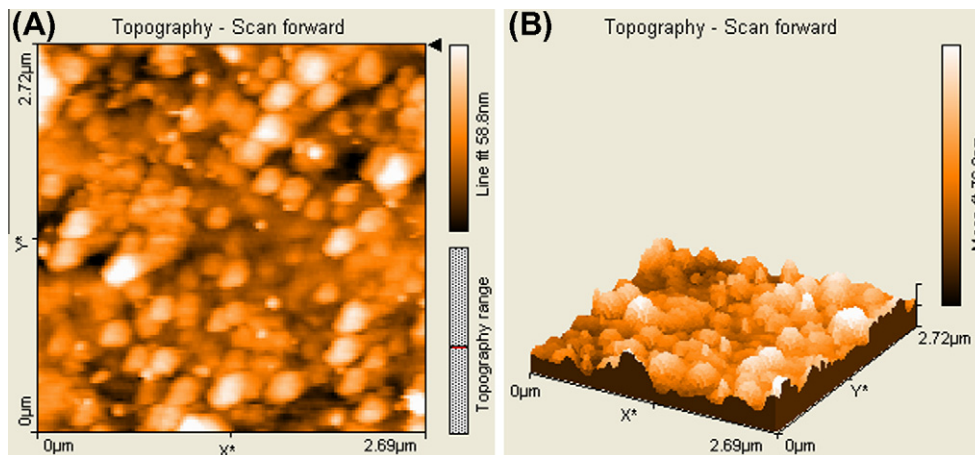


Fig. 4. AFM images of the synthesized TiO₂ NPs showing: (A) increased surface view and (B) 3D view.

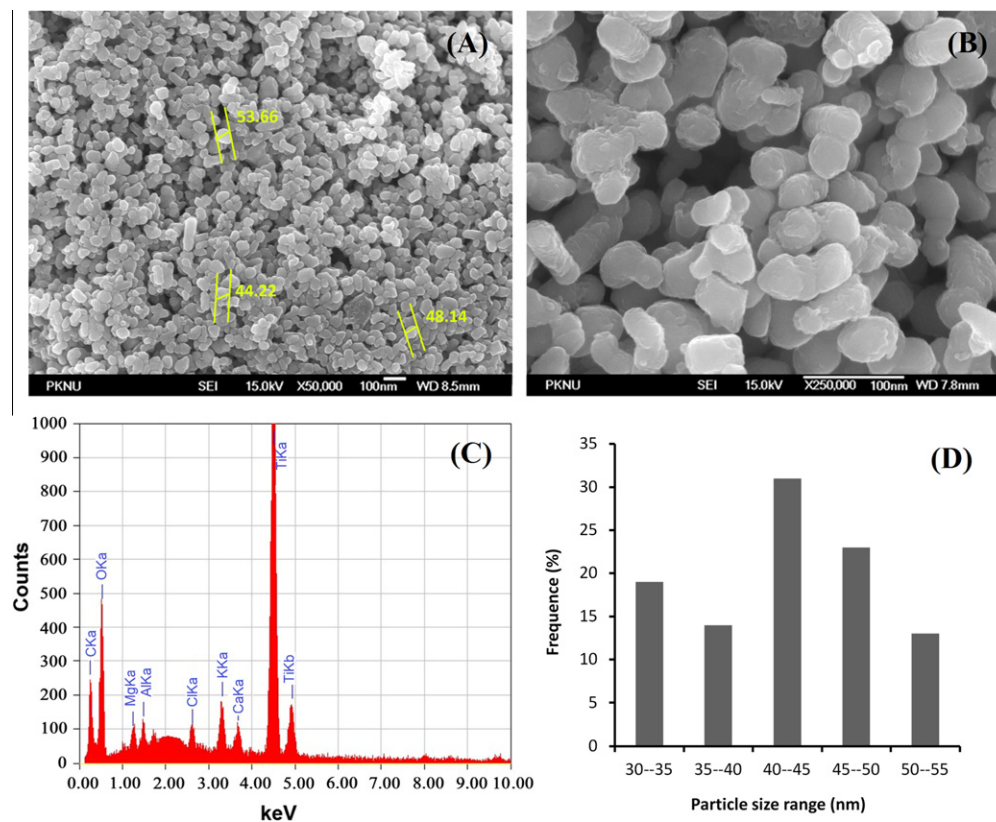


Fig. 5. (A and B) FESEM images of the *A. hydrophila* synthesized TiO_2 nanoparticles, (C) EDX showing chemical composition and (D) particle size distribution.

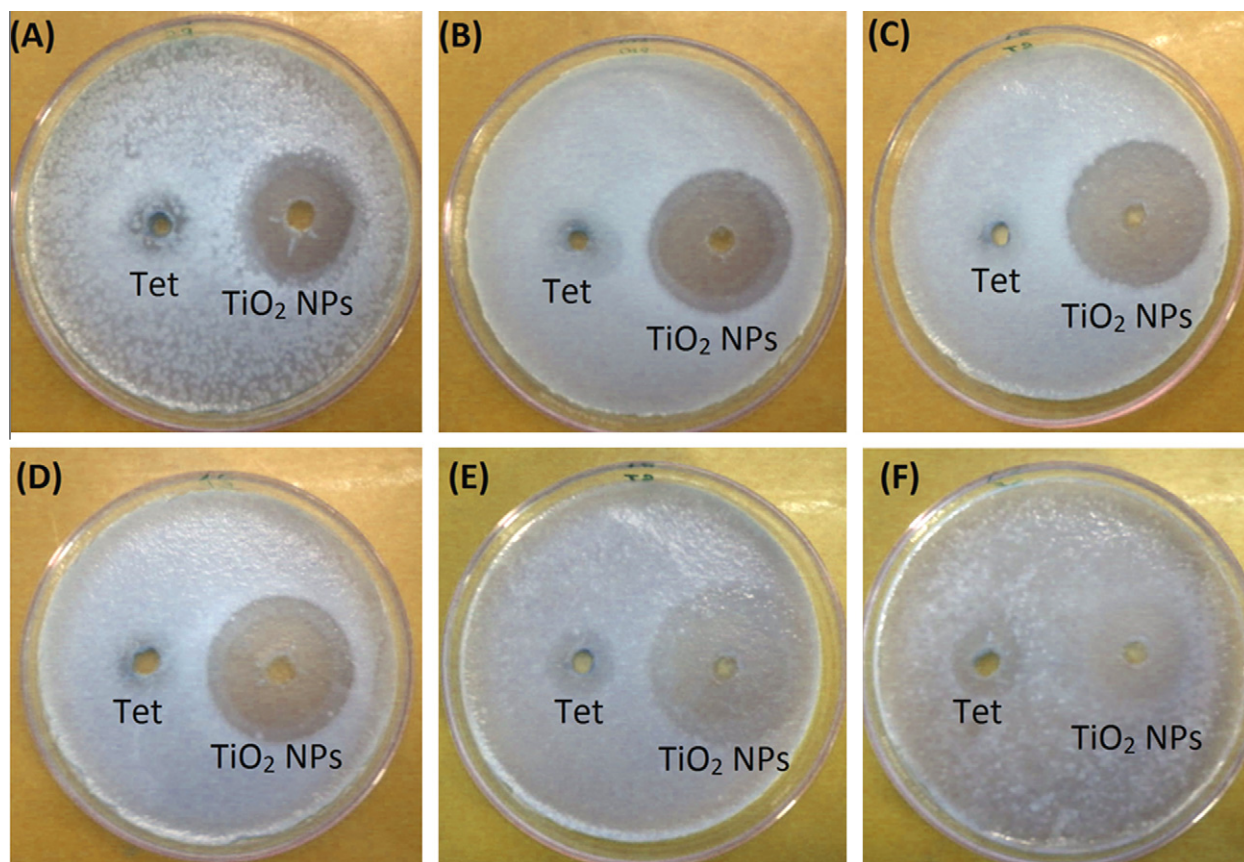


Fig. 6. Well diffusion assay for TiO_2 nanoparticles against *A. hydrophila* (A), *E. coli* (B), *P. aeruginosa* (C), *S. aureus* (D), *S. pyogenes* (E) and *E. faecalis* (F).

Table 2Zone of inhibition (mm) and MIC ($\mu\text{g ml}^{-1}$) of *A. hydrophila* synthesized TiO_2 nanoparticles against various microorganisms.

Microorganism	<i>A. hydrophila</i> synthesized TiO_2 nanoparticles		Tetracycline	
	Zone of inhibition (mm)	MIC ($\mu\text{g ml}^{-1}$)	Zone of inhibition (mm)	MIC ($\mu\text{g ml}^{-1}$)
<i>A. hydrophila</i>	23	25	14	20
<i>E. coli</i>	26	20	14	20
<i>P. aeruginosa</i>	25	30	12	25
<i>S. pyogenes</i>	31	10	15	15
<i>S. aureus</i>	33	10	15	10
<i>E. faecalis</i>	29	15	16	15

When compared to the literature report [35,36], our experimental result supports for the antibacterial activity of tetracycline.

The MIC observed in the present study for synthesized TiO_2 NPs were $25 \mu\text{g ml}^{-1}$ for *A. hydrophila*, $20 \mu\text{g ml}^{-1}$ for *E. coli*, $30 \mu\text{g ml}^{-1}$ for *P. aeruginosa*, $10 \mu\text{g ml}^{-1}$ for *S. pyogenes*, $10 \mu\text{g ml}^{-1}$ for *S. aureus*, and $15 \mu\text{g ml}^{-1}$ for *E. faecalis* (Table 2). The bactericidal effect of TiO_2 generally has been attributed to the decomposition of bacterial outer membranes by reactive oxygen species (ROS), primarily hydroxyl radicals (OH $\cdot$), which leads to phospholipid peroxidation and ultimately cell death [37,38]. It was proposed that nanomaterials that can physically attach to a cell can be bactericidal if they come into contact with cell [39]. If the membrane of a bacterium is compromised, the cell may repair itself or, if the scratch is severe, the cell component may be released and eventually the cell will die [40].

Control of pathogenic microbes by antimicrobial NP was a promising approach to defeat the multiresistant pathogens such as methicillin-resistant *S. aureus* [41]. There are several possible mechanisms to explain the bactericidal effect of TiO_2 particles. TiO_2 exhibits antimicrobial activity due to its strong oxidizing property when exposed to sunlight or ultraviolet light. The microbial surface was the primary target of the initial oxidative attack when irradiated TiO_2 particles come into contact with microbes. ROS, such as hydrogen peroxide, hydroxyl radical and superoxide anions, generated on the irradiated TiO_2 surface initially promotes peroxidation of the polyunsaturated phospholipids component of the lipid of the microorganism [24]. Kuhn et al. [42] described that light and scanning electron microscopic examinations suggest the microbial destruction achieved takes place through direct damage to cell walls caused by hydroxyl radical (OH $\cdot$). For the gram-positive cell walls such as for *S. aureus*, normally thick, homogeneous cell wall was composed primarily of peptidoglycan, which accounts for nearly 80% of their cell wall. The rest was 10–20% of teichoic acids and other such as proteins and lipopolysaccharides [43]. Gram-negative cell walls were much more complex than gram-positive walls. This result showed that TiO_2 nanoparticles were effective for inhibiting both gram-positive and gram-negative bacteria.

Conclusion

We report a simple biological method for the synthesis of TiO_2 NPs using the bacteria *A. hydrophila* and its antibacterial activity. The compounds found in *A. hydrophila* were uric acid, glycyl-L-glutamic acid, glycyl-L-proline and L-Leucyl-D-leucine. The major secondary metabolite which is present in the culture broth is glycyl-L-proline which may be responsible for the synthesis of the TiO_2 NPs. The FTIR spectrum of the synthesized TiO_2 NPs were 3430, 2937, 1643, 1403 and 1079 cm^{-1} and the results indicated that alcohols, phenols, primary amines, lactones and aliphatic amines in *A. hydrophila* may have been participated in the process of nanoparticle synthesis. The crystallite sizes were calculated using Scherrer's formula applied to the major intense peaks

and found to be in the range of 40.50 nm. This method is considered as an innovative approach for the synthesis of TiO_2 NPs possessing significant antibacterial activity.

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